

AI61002 : Deep Learning Foundations and Applications

Spring 2026

Assignment 4

(20 Marks)

Instructions: Complete the tasks listed below, ensuring you follow the specified instructions for each. Provide step-by-step outputs in the Jupyter Notebook (.ipynb file). All relevant code snippets, visualizations, and explanations must be included within the notebook. Submit the required files as per the submission guidelines provided.

Question 1 - 2 marks

Load the Flickr8K image captioning dataset from Huggingface datasets: <https://huggingface.co/datasets/jxie/flickr8k> and show some samples from the dataset and corresponding captions. (Note: there are 5 captions corresponding to each image).

Question 2 - 1 marks

Split the dataset into training (6000), validation (1000), and test partitions (1000), ensuring that each partition contains strictly distinct images.

Question 3 - 5 marks

Design a transformer encoder-decoder architecture from scratch as done in the class tutorial (i.e., using `nn.MultiheadAttention`, `nn.Linear`, and `nn.LayerNorm` with residual connections). The encoder should take the images as inputs, and the decoder should generate the text caption in an autoregressive fashion. Use a linear embedding layer (`nn.Embedding`) and sinusoidal positional encoding. Choose suitable values of the hyperparameters such as the embedding dimension, the number of attention heads, the number of encoder and decoder layers to get a good fit. Show the model architecture and report the number of trainable parameters.

Question 4 - 1 marks

Design a suitable loss function for the autoregressive image captioning task.

Question 5 - 3 marks

Train the custom transformer model using the Adam optimizer and show the variation of training and validation losses against the number of epochs.

Question 6 - 2 marks

Evaluate the transformer model on the test set and report the performance by computing the BLEU score of the generated caption against each of the 5 reference captions. Report the average BLEU score over the entire test dataset.

Question 7 - 3 marks

Now replace the custom transformer encoder and decoder modules in your model with PyTorch's native `nn.TransformerEncoderLayer()` and `nn.TransformerDecoderLayer()` and train the model again. Show the variation of training and validation losses. (Note: use the same hyperparameters such embedding dimension, number of encoder and decoder layers, number of attention heads, and sizes of the positionwise feedforward neural network as used for the custom transformer model). Show the model architecture and report the number of trainable parameters.

Question 8 - 1 marks

Report the performance of the trained model using the native `nn.TransformerEncoderLayer()` and `nn.TransformerDecoderLayer()` on the test set by computing the average BLEU score (note each generated caption should be compared with all 5 captions in the dataset and the corresponding bleu scores averaged). .

Question 9 - 2 marks

Import the pre-trained ViT-GPT2 model from Huggingface (<https://huggingface.co/nlpconnect/vit-gpt2-image-captioning>) and get the image captions generated by this model on the test set. Report the average BLEU score on the dataset in the same manner as for the trained transformer models and briefly compare the results.

Submission Guidelines

- The content that you submit must be your individual work.

- Submit both .py and .ipynb file formats. Both submissions are mandatory for credit.
- Name the files as "<roll_number>_assignment_4.py" and "<roll_number>_assignment_4.ipynb". Submit them in a single zip file named "<roll_number>_assignment_4.zip". For example, if your roll number is 23AI91R01, the submission file should be named 23AI91R01_assignment_4.zip (Note: do not submit .rar files).
- Ensure all code is well-commented and explanations are included using text cells in the Jupyter Notebook.
- Submit your assignment through Moodle before the deadline. Late submissions or email submissions will not be accepted.

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